IEOR 221

Homework 3

Module 4 (September 22, 2025)

1) A $100 million interest rate swap has a remaining life of 10 months. Under the terms of the swap, six-month LIBOR is exchanged for 4% per annum (compounded semiannually). Six-month LIBOR forward rates for all maturities are 3% (with semiannual compounding). The six-month LIBOR rate was 2.4% two months ago. OIS rates for all maturities are 2.7% with continuous compounding. What is the current value of the swap to the party paying floating? What is the value to the party paying fixed? (Hull 7.12)

2) A currency swap has a remaining life of 15 months. It involves exchanging interest at 10% on £20 million for interest at 6% on $30 million once a year. The term structure of risk-free interest rates in the United Kingdom is flat at 7% and the term structure of riskfree interest rates in the United States is flat at 4% (both with annual compounding). The current exchange rate (dollars per pound sterling) is 1.5500. What is the value of the swap to the party paying sterling? What is the value of the swap to the party paying dollars? (Hull 7.14)

3) A financial institution has entered into an interest rate swap with company X. Under the terms of the swap, it receives 4% per annum and pays six-month LIBOR on a principal of $10 million for five years. Payments are made every six months. Suppose that company X defaults on the sixth payment date (end of year 3) when six-month forward LIBOR rates for all maturities are 2% per annum. What is the loss to the financial institution? Assume that six-month LIBOR was 3% per annum halfway through year 3 and that at the time of the default all OIS rates are 1.8% per annum. OIS rates are expressed with continuous compounding; other rates are expressed with semiannual compounding. (Hull 7.19)

4) OIS rates are 3.4% for all maturities. What is the value of an OIS swap with two years to maturity where 3% is received and the floating reference rate is paid. Assume annual compounding, annual payments, and $100 million principal. (Hull 7.27)